Ayush Chaudhary

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PROFESSIONAL SUMMARY

Backend developer with hands-on experience in Python, FastAPI, Django, Flask, PostgreSQL, Redis and Docker. Experienced in building REST APIs, optimizing slow queries through indexing and caching, and delivering containerized projects with CI/CD. Has also built multi-agent and RAG systems on Ollama and led a research project on digital payments and consumer spending. Seeking a backend role building features and APIs within a product and engineering team.

EDUCATION

GITAM Deemed University, B.Tech in Computer Science 2022 - 2026

EXPERIENCE

Intern, Data Engineering & ML Infrastructure, Labmentix May 2025 - Jul 2025

- Developed Python scripts using Pandas and NumPy to automate categorical encoding, normalization and scaling, removing manual dataset preparation.
- Implemented validation checks for schema mismatches, incorrect data types and missing values ahead of model training.
- Set up monitoring to flag data inconsistencies, redundant features and drift.

Machine Learning Intern, Backend & Data Systems, CodeBeat Placement May 2024 - Aug 2024

- Built and deployed a Flask REST API serving live model predictions, with request validation, error handling and connection pooling. The API powered a Streamlit dashboard used for daily decisions.
- Optimized slow PostgreSQL queries by removing N+1 patterns and adding indexes, and introduced a Redis cache for pipelines processing large daily record volumes.
- Reduced dashboard load time by caching query results in Redis and limiting the data sent to the frontend.

Research Intern (Team Lead), IISPR 8-member team

- Led an 8-member team researching the effect of digital payments on consumer spending in India, including a review of the literature on UPI adoption, spending psychology and merchant digitisation, for the paper "From Cash to UPI: Reconfiguring Consumer Spending Behaviour in India, 2016–2025."
- Analysed NPCI and RBI UPI data for FY2016–17 to FY2024–25, covering average transaction size, year-on-year growth and the P2P/P2M split, to distinguish payment substitution from genuine changes in spending behaviour.

PROJECTS

Offline Code Compiler and Secure Contest System [Electron, Node.js, AES-256, Docker]

- Led a team of 4 in building an Electron and Node.js/Express desktop app that compiles and runs C++, Java and Python offline, with Monaco Editor for code input.
- Released a Docker version with a CI/CD pipeline, used in 2 college hackathons.
- Built the LAN contest platform with automatic test case checking and feedback on compile and runtime errors.
- Encrypted test cases and submissions with AES-256-CBC and implemented password-protected login with session management.

LocalOPS, Multi-Agent AI Inference System [FastAPI, Ollama, Redis, Python]

- Built a multi-agent system behind a FastAPI REST API, with a router agent handling concurrent requests, Redis for state and async request batching.
- Improved throughput with streaming responses, cached quantized GGUF models and RAG-based memory. Profiled the system and resolved latency bottlenecks on a GTX 1650 (4GB VRAM).
- Added error handling, request validation and timeouts, and containerized the system with Docker.

SKILLS

Languages: Python, SQL, JavaScript (Node.js)
Backend & APIs: FastAPI, Django, Flask, Express.js, REST API design, authentication and authorization
Databases: PostgreSQL, MySQL, SQLite, Redis, query optimization, relational database design
DevOps & Tools: Docker, CI/CD, Electron, Streamlit
Security & Reliability: AES-256-CBC encryption, session management, request validation, error handling
Data & Machine Learning: Pandas, NumPy, Scikit-learn, LSTM, time-series forecasting, feature engineering, sentiment analysis
AI & LLM Systems: Ollama, multi-agent systems, RAG, model quantization, prompt engineering
Research: Secondary-data analysis, literature review, academic writing